

SignalHarbor

Explainable text risk signals
and portfolio stress testing

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AI/NLP Risk Engine + Module B

Problem and approach

A reviewable path from an unstructured report to a portfolio scenario

News and social posts contain duplicates, uncertain claims, and routine updates. A negative headline alone does not identify an exposure.

SignalHarbor retains the source, extracts a structured event, flags weak evidence, and applies explicit assumptions to a synthetic banking portfolio.

EXAMPLE STRUCTURED SIGNAL

event: "Geopolitical"

sentiment: -0.96

impact: 10 / 10

source_type: "news"

synthetic: true

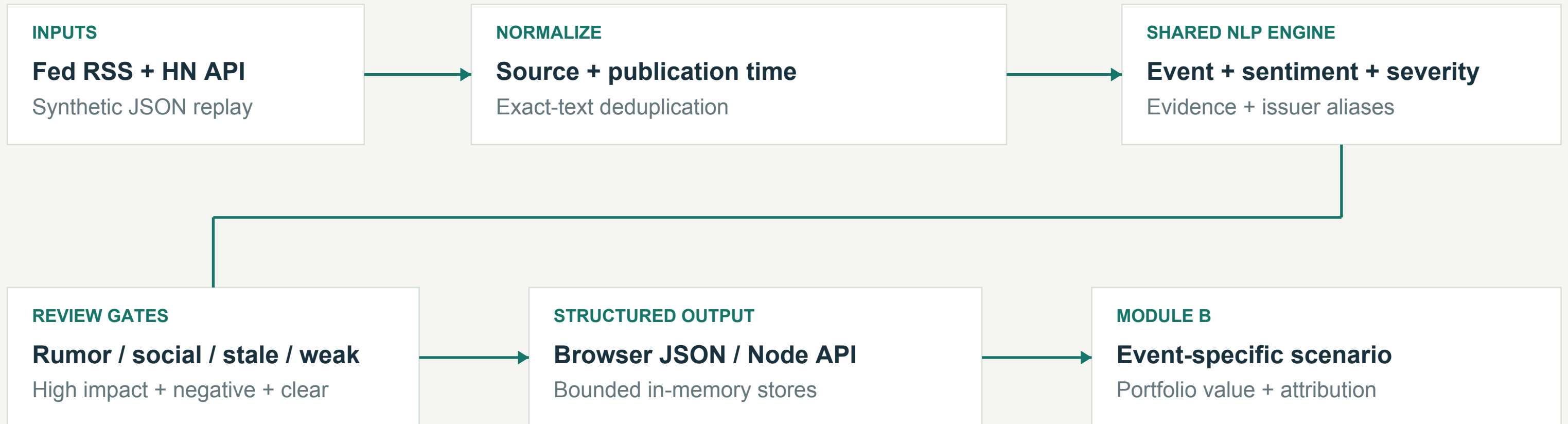
review_required: false

stress_triggered: true

Example: data/news.json, n1. Scores and scenarios are prototype outputs, not predictions.

System design

One engine serves the browser and Node API; the same signals feed the stress lab



Live adapters: federalreserve.gov/feeds/feeds.htm and hn.algolia.com/api. Source failures remain visible.

Implementation and review gates

Small, inspectable components make every output reproducible

Event model

84 synthetic training examples; seven classes.

Naive Bayes with word and two-word features; alpha 0.5.

Sentiment

Financial lexicon with local negation.

A bounded score in $[-1, 1]$; evidence terms remain attached.

Severity

Event base, sentiment magnitude, and severe terms.

A transparent 1-10 proxy, not a trained loss forecast.

Review gate

Rumors, social posts, stale reports, and weak evidence.

Automatic trigger requires impact > 7 and negative sentiment.

Reference: scikit-learn.org/stable/modules/naive_bayes.html. Classifier scores are uncalibrated.

Measured prototype results

All accuracy claims below refer only to the supplied synthetic diagnostic set

28 / 28

Event classifications matched

100% accuracy; macro F1 1.000.
Majority-class baseline: 14.3%.

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Functional and HTTP tests passed

Input errors, review gates, source parsing,
issuer scope, hedge sign, and reconciliation.

0.407 ms

Local engine p95 across 1,000 calls.
Node 23.11.0 / macOS arm64. Network, training, and UI excluded.

The evaluation has 28 separately authored sentences from the same author and domain as training.
Real-news accuracy, sentiment accuracy, and severity calibration remain unmeasured.

Evidence: data/evaluation_results.json and npm test. Timing is a measured sample; reruns vary.

Portfolio impact with attribution

Illustrative geopolitical scenario at severity 10; all values in USD millions

100.00

Baseline

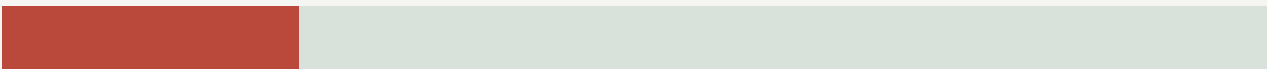
92.32

Stressed value

-7.68

Scenario P&L (-7.68%)

Rates



-0.675

Spreads



-2.325

Equity



-2.880

Loan credit



-1.800

Assumptions

+75 bps rates

+150 bps spreads

-12% equity

4% loan haircut

The pay-fixed swap gains \$0.4875m within the net rate effect. Each scenario starts from the same \$100m baseline.

Synthetic portfolio: 11 positions. First-order sensitivities; no probabilities or market forecasts.

Limitations and next steps

A transparent baseline for a live jury discussion

Language and confidence

Validate on independently labeled financial news with a time-based split; calibrate uncertainty.

Source and entity matching

Cluster paraphrases, reconcile independent reports, and replace fictional aliases with validated entity links.

Financial realism

Use instrument repricing and validated default, liquidity, and nonlinear risk models.

Operational deployment

Add durable storage, authenticated APIs, feed monitoring, and audit trails before shared use.

AI assistance supported the original code, synthetic examples, tests, and documentation. The participant must understand and defend the implementation.